

Evidence-Grounded Quarterly Performance Reporting

Architecture and the AI boundary

The system takes only client code and quarter, discovers sources by content, and builds one report model for PDF rendering and local evidence review. Python owns workbook extraction, financial calculations, chronology, fund selection, ranking and compliance. PyMuPDF handles all PDF extraction and the original flash appendix; openpyxl, ReportLab and Matplotlib keep the runtime portable without Office, OCR executables or downloaded document models. The Windows launcher bootstraps pinned dependencies with one command.

One bounded evidence agent chooses read-only tools: list, scoped search, inspect a registered passage, validate an exact quote, finish or escalate. Each action receives an observation and validation feedback before the next decision. It has an eight-action limit, repeated-action stop, and shared API budget; it cannot open arbitrary files, alter numbers or approve reports. Its investigated claims and trace guide human inspection, while deterministic heading-delimited excerpts keep the report's numerical inventory independent of its focus choices. This is intentionally a hybrid agentic workflow, not an autonomous investment analyst. No orchestration framework or separate RAG database is needed for this small source set; a retrieval boundary permits later integration with a shared company search service.

Verification and regression detection

The other model job is chart-image extraction through the approved OpenAI/Anthropic APIs. Labeled values are transcribed; unlabeled bars/line points are approximate gridline readings at declared precision. Mixed-confidence charts are conservatively approximate throughout. The schema checks labels, units, duplicates, finite values and model-declared axis bounds. A human must confirm every extraction. Transaction/appraisal spreads and their direction are calculated in Python. Financial connecting prose is templated; manager text is source-extractive. There is no free-form model prose capable of inserting unsupported financial figures.

All authoritative money arithmetic uses Decimal; currency identities have zero tolerance, and fractions of a cent, booleans and nonfinite values are rejected. JSON preserves currency as base-ten strings. Rounding is a presentation boundary, never a source overwrite. Contribution is income plus appreciation less fees; flows belong only in roll-forward checks. Rankings retain overlapping roles; TWR/IRR and weighted LTV are read, not recomputed. Rounded manager statistics are separate, precision-aware corroboration checks. IC/one-pager amounts must match exactly; history is checked at its displayed USD-million precision and its missing client column is disclosed.

Golden and adversarial tests cover one-cent errors, shifted/renamed inputs, added holdings, duplicate sources, invalid IC actions/dates, fabricated quotes, agent limits, chart schema failures and approval tampering. Ten fixed-seed randomized cases permute discovery/IC inputs or corrupt selected cash values; these are constrained tests, not unrestricted fuzzing. The regression matrix exports observed results. Two independent no-key runs and two live API runs with empty caches passed canonical comparison: financial/chart figures, fund selection/order/role, compliance and structure. Diagnostic wording differed. Temperature zero, a snapshot model, seed and content cache help but do not guarantee provider determinism; semantic differences fail the test.

Human review and finalization

Clicking report blocks opens locators and recursive calculation inputs. The review queue distinguishes warnings, blockers and required chart confirmations. Meridian's missing report permits flash-only commentary, never invented asset drivers. Missing policy, ambiguous required sources, currency reconciliation failures and unavailable chart readings prevent finalization. Existing automatic chart readings can be corrected, but each correction must pass the same schema and include a reviewer reason. Money corrections belong in the authoritative workbook followed by regeneration.

Backend approval requires a named acknowledgment, every chart confirmation, no blockers, and unchanged draft, source and exhibit/asset hashes. It creates a final PDF plus hash-bound approval, evidence and semantic manifests. Regeneration invalidates old approved artifacts. This guards accidental stale approval, not a hostile filesystem owner or authenticated multi-user release process. No-key mode retains complete deterministic financial output, source excerpts and all report sections; the agent is disabled and image readings are unavailable, so it remains a flagged draft. The supplied live draft still requires the applicant's review/approval.

Next quarter and format changes

A full-pipeline rehearsal generates a different client, 1Q26 sources, renamed files, moved rows, one removed and two added holdings, and an approval carried from a 2025 log into a 2026 withdrawal. It completes without production edits; its 13 holdings, NAV and surviving approval match fixture expectations. This no-key rehearsal tests adaptation, not live vision accuracy. Discovery records accepted/ignored/failed files and reasons. Tables use labels/grouped headers, not coordinates. Canonical entity rules normalize punctuation, legal suffixes and Roman numerals; collisions and ambiguous matches are not guessed.

Client identity and benchmark come from the flash; policy comes from the prior PMR. The prior PDF supplies typography, primary color and contents order; inspected table/chart tokens are reused. Blocks flow as holdings or commentary grow. Unsupported new policy syntax blocks release; new section structure is flagged. Truly different financial measures or arbitrary redesign require an adapter, not a claim of universal document understanding.

Specific judgments and remaining work

Traceability is a machine-readable claim manifest plus fact ledger: file, sheet/cell, page or slide/chart, raw values, numeric metadata, formulas and input IDs. Attribution cells have individual references; other blocks have scoped evidence references. Run diagnostics expose stage durations, calls/usage, cache hits/misses, failures and the agent timeline without hidden reasoning or credentials. All outputs remain inspectable without a monitoring service.

The SPEC-designated FundingStatus/AnnReturns rows own portfolio summary/compliance; Vectera Initiated Investments owns explicitly labeled funded-program inception statistics. The prior attribution table omits a selected individual position: this table includes all required roles. The sample's 'Of which funded' figure is a commitment amount; this report labels it 'Flash commitment total.' The realized-only \$0.51 discrepancy is outside the active portfolio and is a warning. Broad market returns are not conflated with the client benchmark; mandate appointment dates and performance inception are not assumed to mean the same thing. The appendix preserves source content as vector panels on Letter paper, repeating investment names when wide tables split; the original six-page PDF is also retained.

Live OpenAI access and both fresh-run comparisons have been observed on macOS/Python 3.12. Windows 11 bootstrap/interactive execution and final human approval remain outstanding, and are not claimed as verified. With more time: authenticated approvals, broader scanned-document/policy adapters, deterministic chart measurement, and more cross-client fixtures. I deliberately omitted Terraform, Docker dependence, multiple agents and general paraphrasing to keep the submission portable, bounded and defensible.